

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. **Name of the Academic Unit:** Civil Engineering
2. **Subject Name:** Water Resources Engineering **L-T-P:** 4-0-0 **Credits:** 4
3. **Pre-requisites:** Hydraulics
4. **Syllabus and reference books:**

Syllabus:

Introduction: Hydrologic Cycle and abstractions of precipitation, Infiltration and Evapotranspiration: Forms of precipitation, measurement, depth-duration and intensity duration frequency relations. Evaporation – process, measurement, and estimation, Infiltration process, measurement and estimation. Evapotranspiration measurement and estimation. Runoff and Hydrographs: Rainfall Runoff correlations, Flow duration curve, Mass curve, Droughts and floods, Factors affecting flow hydrograph. Unit hydrograph, its analysis and S-curve hydrograph, Synthetic and instantaneous unit hydrographs. Statistical analysis: Risk, reliability, and safety factor, Flood frequency studies. Rational method, Time Area curves; Design flood; Channel and flood routing. Groundwater and well hydrology: Flow equations, Confined and unconfined flow; Well hydraulics; Steady and unsteady flow, Well losses, Specific capacity. River basins and watershed management: Water resources - storages and their yields; Water logging; Streams and their gauging; River morphology; Types of floods and drought, and their management; Capacity of reservoirs. Irrigation: types of irrigation systems, Soil-water plant relationship: Classification of soil water- soil moisture contents- depth of soil water available to plants; root zone soil water; permanent and ultimate wilting point. Depth of water applied during irrigation- Duty of water and delta improvement of duty, command area and intensity of irrigation consumptive use of water and Evapotranspiration irrigation efficiencies- assessment of irrigation water, frequency of irrigation. Canal systems - alignment of canals, canal losses, Design of channels - channels carrying clear and sediment laden water, alluvial channels carrying clear and sediment laden water, Lacey's theory of regime channels. Canal outlets: non-modular, semi-modular and modular outlets. Water logging: causes, effects and remedial measures. Lining of canals economics of lining, types of lining. Drainage of irrigated lands: necessity, methods. Canal structures- Surface and sub-surface flow considerations for design of canal structures: hydraulic jump, seepage forces, uplift forces. Silt exclusion devices, Canal falls, cross regulator, distributary head regulator. Cross drainage works: their classification and their application areas; Canal falls and escapes. Gravity dams: cross section of overflow and spillway blocks; ogee spillway and energy dissipaters. Forces on gravity dams, Stability analysis; causes of failure, stress analysis. Water diversion structures: Design of weirs and barrages on permeable foundation; Different units of head works, types of weirs, sediment control in canals, river training for canal head works. Theories of seepage for design of weirs: Khosla-Bose-Taylor method of independent variables.

Reference Books:

1. Engineering Hydrology by K. Subramanya. McGraw Hill Education
2. Irrigation and Water Resources Engineering by G.L. Asawa, New Age International Pvt Ltd
3. Hydrology and Water Resources Engineering Vol.1 by S. K. Garg. Khanna Publishers.
4. Water Resources Engineering, Web-resource – NPTEL. (<https://nptel.ac.in/courses/105/105/105105110/>)

5. Lecture-wise break-up:

Sl. No.	Topic	No. of lectures
1.	Introduction: Hydrologic Cycle and abstractions of precipitation, Infiltration and Evapotranspiration: Forms of precipitation, measurement, depth-duration and intensity duration frequency relations	4
2.	Evaporation – process, measurement, and estimation, Infiltration process, measurement and estimation. Evapotranspiration measurement and estimation	4
3.	Runoff and Hydrographs: Rainfall Runoff correlations, Flow duration curve, Mass curve, Droughts and floods, Factors affecting flow hydrograph. Unit hydrograph, its analysis and S-curve hydrograph, Synthetic and instantaneous unit hydrographs	4
4.	Statistical analysis: Risk, reliability, and safety factor, Flood frequency studies. Rational method, Time Area curves; Design flood; Channel and flood routing	4
5	Groundwater and well hydrology: Flow equations, Confined and unconfined flow; Well hydraulics; Steady and unsteady flow, Well losses, Specific capacity	4
6	River basins and watershed management: Water resources - storages and their yields; Water logging; Streams and their gauging; River morphology; Types of floods and drought, and their management; Capacity of reservoirs.	4
7	Irrigation: types of irrigation systems, Soil-water plant relationship: Classification of soil water- soil moisture contents- depth of soil water available to plants; root zone soil water; permanent and ultimate wilting point	4
8	Depth of water applied during irrigation- Duty of water and delta improvement of duty, command area and intensity of irrigation consumptive use of water and Evapotranspiration irrigation efficiencies-	4

	assessment of irrigation water, frequency of irrigation.	
9	Canal systems - alignment of canals, canal losses, Design of channels - channels carrying clear and sediment laden water, alluvial channels carrying clear and sediment laden water, Lacey's theory of regime channels. Canal outlets: non-modular, semi-modular and modular outlets. Water logging: causes, effects and remedial measures. Lining of canals economics of lining, types of lining. Drainage of irrigated lands: necessity, methods	4
10	Canal structures- Surface and sub-surface flow considerations for design of canal structures: hydraulic jump, seepage forces, uplift forces. Silt exclusion devices, Canal falls, cross regulator, distributary head regulator. Cross drainage works: their classification and their application areas; Canal falls and escapes	4
11	Gravity dams: cross section of overflow and spillway blocks; ogee spillway and energy dissipaters. Forces on gravity dams, Stability analysis; causes of failure, stress analysis.	4
12	Water diversion structures: Design of weirs and barrages on permeable foundation; Different units of head works, types of weirs, sediment control in canals, river training for canal head works. Theories of seepage for design of weirs: Khosla-Bose-Taylor method of independent variables.	4
Total number of hours		48